Optionlytics “Lite”

Web version of the application (OR)

Description

- You are to build a “lite” version of OR/Optionlytics for the web
- You will need to use JavaScript and an assortment of framework to build the website
- You will need to setup the infrastructure (NodeJS) on a server for us (we will provision for you)

Requirements

- Web based
- Load and responsive rate must be fast
- Must be multi-threaded and asynchronous to handle data streams and updating data calculations
- User must be allowed to have multiple windows open

Infrastructure

- MySQL Database
- NodeJS
- Electron Framework
- www.Photonkit.com/

Commodities Supported

- Henry Hub (NG)
- WTI (CL)
- WTI CMA (tbd)
- RBOB (RB)
- Heating Oil (HO)
- Brent Oil (BB)

Website Sections

- Login screen (this should be the primary screen you see when you go to the website – this is just a username, password, forgot password)
- Public cannot create usernames

<u>When logged into website</u>

- Option Grid
  - Search (search box appears)
- Pricer (Options)
- Futures
- User Profile
- Admin Screen

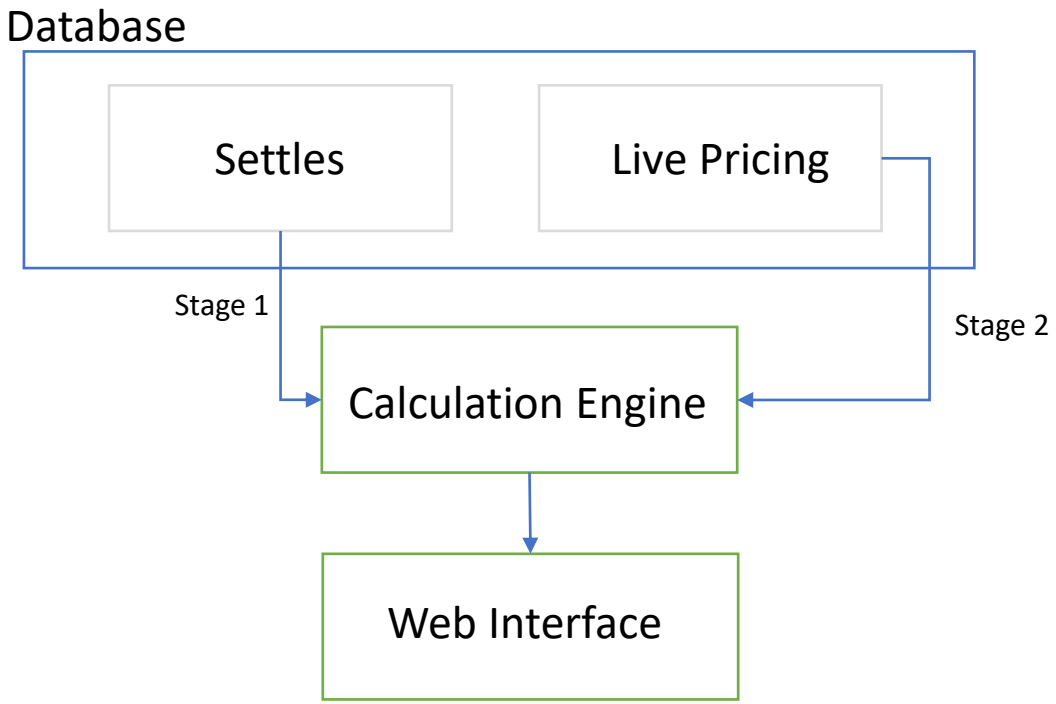
Website Comments

- Option Grid is the “home screen” once logged in.

- We have not fully worked out the web GUI interface, but we would like something minimal (white, light blue, gray, black)

Settles vs. Live Streaming

- Going to start phase one just using settlement pricing
- Phase two we are going to add a table which streams prices into the table
- This web application must be able to calculate on the fly the management of Live pricing and updating all the required information.



Black (JavaScript)

- There are a couple libraries online Black76/Black Scholes option pricing

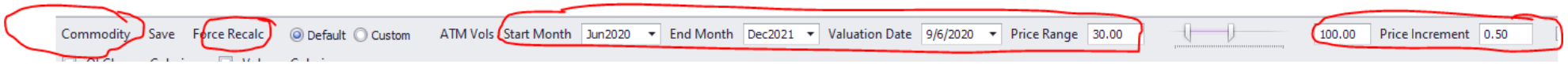
- https://github.com/vollib

- Alternatively, you can convert our C# (just the black() functionality.

Option Grid – Core Features

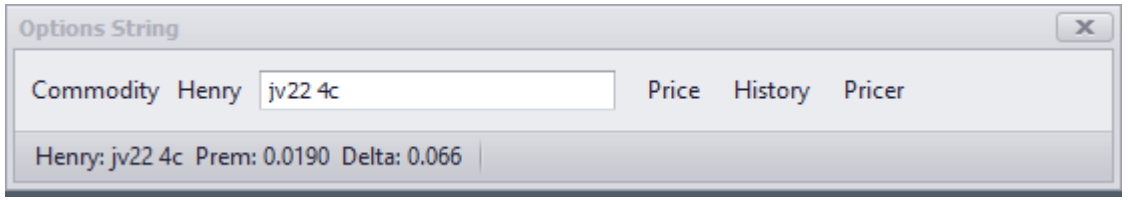
- By commodity (drop down menu)
- Add months
- Add strike range (min / max)
- Add increments
- Required data:
  - Put Premium Settle
  - Call Premium Settle
  - Put Premium (current)
  - Call Premium (current)
  - Implied volatility
  - Volatility skew
  - Vol Change
  - GammaBE
  - Delta
  - Gamma
  - Vega
  - Theta
- We would use the same color coding here

	Oct20(V20) | $39.77 | 0.00									Nov20(X20) | $40.15 | 0.00									Dec20(Z20) | $40.59 | 0.00									Jan21(F21) | $41.02 | 0.00									Feb21(G21) | $41.44 | 0.00					
	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpV
30.000	9.850		30.000	0.083	0.080	2.194	0.876	0.876	0.455	10.440		30.000	0.301	0.290	1.644	0.650	0.650	0.250	11.170		30.000	0.601	0.590	1.531	0.599	0.599	0.175	11.820		30.000	0.831	0.800	1.486	0.575	0.575	0.151	12.400		30.000	1.001	0.960	1.440	0.552	0.5
30.500	9.360		30.500	0.094	0.090	2.133	0.851	0.851	0.430	9.970		30.500	0.330	0.320	1.609	0.636	0.636	0.237	10.720		30.500	0.650	0.630	1.506	0.589	0.589	0.165	11.380		30.500	0.890	0.850	1.463	0.566	0.566	0.142	11.970		30.500	1.069	1.030	1.421	0.544	0.5
31.000	8.870		31.000	0.104	0.100	2.064	0.824	0.824	0.403	9.490		31.000	0.350	0.350	1.559	0.616	0.616	0.217	10.270		31.000	0.700	0.680	1.479	0.578	0.578	0.154	10.940		31.000	0.948	0.930	1.438	0.557	0.557	0.132	11.550		31.000	1.147	1.110	1.405	0.538	0.5
31.500	8.380		31.500	0.113	0.110	1.988	0.793	0.793	0.372	9.030		31.500	0.390	0.380	1.532	0.606	0.606	0.206	9.820		31.500	0.748	0.740	1.448	0.566	0.566	0.143	10.510		31.500	1.017	1.000	1.417	0.549	0.549	0.124	11.130		31.500	1.225	1.190	1.387	0.531	0.5
32.000	7.890		32.000	0.122	0.120	1.911	0.763	0.763	0.342	8.560		32.000	0.419	0.410	1.488	0.588	0.588	0.188	9.380		32.000	0.808	0.800	1.423	0.557	0.557	0.133	10.090		32.000	1.094	1.070	1.399	0.541	0.541	0.117	10.720		32.000	1.314	1.280	1.372	0.526	0.5
32.500	7.400		32.500	0.133	0.130	1.835	0.732	0.732	0.311	8.100		32.500	0.459	0.450	1.453	0.575	0.575	0.175	8.950		32.500	0.876	0.860	1.401	0.548	0.548	0.124	9.670		32.500	1.174	1.150	1.379	0.534	0.534	0.109	10.310		32.500	1.403	1.370	1.355	0.519	0.5
33.000	6.920		33.000	0.153	0.150	1.778	0.710	0.710	0.288	7.640		33.000	0.498	0.490	1.413	0.559	0.559	0.159	8.520		33.000	0.946	0.930	1.376	0.538	0.538	0.114	9.260		33.000	1.262	1.240	1.361	0.527	0.527	0.103	9.910		33.000	1.500	1.470	1.340	0.513	0.5
33.500	6.440		33.500	0.171	0.170	1.713	0.684	0.684	0.262	7.190		33.500	0.547	0.540	1.380	0.546	0.546	0.146	8.100		33.500	1.024	1.010	1.354	0.529	0.529	0.106	8.850		33.500	1.352	1.330	1.342	0.519	0.519	0.095	9.520		33.500	1.608	1.580	1.326	0.508	0.5
34.000	5.960		34.000	0.192	0.190	1.644	0.656	0.656	0.235	6.740		34.000	0.597	0.590	1.343	0.531	0.531	0.131	7.690		34.000	1.113	1.100	1.334	0.522	0.522	0.098	8.450		34.000	1.449	1.430	1.323	0.512	0.512	0.088	9.120		34.000	1.706	1.680	1.308	0.501	0.5
34.500	5.480		34.500	0.211	0.210	1.567	0.626	0.626	0.204	6.310		34.500	0.666	0.660	1.317	0.521	0.521	0.121	7.280		34.500	1.202	1.190	1.312	0.513	0.513	0.089	8.060		34.500	1.558	1.540	1.307	0.506	0.506	0.082	8.740		34.500	1.826	1.800	1.295	0.496	0.4
35.000	5.020		35.000	0.252	0.250	1.525	0.609	0.609	0.188	5.880		35.000	0.736	0.730	1.286	0.509	0.509	0.109	6.880		35.000	1.300	1.290	1.290	0.505	0.505	0.081	7.670		35.000	1.666	1.650	1.288	0.499	0.499	0.074	8.360		35.000	1.943	1.920	1.279	0.490	0.4
35.500	4.550		35.500	0.281	0.280	1.453	0.580	0.580	0.159	5.450		35.500	0.805	0.800	1.250	0.494	0.494	0.095	6.490		35.500	1.409	1.400	1.271	0.497	0.497	0.073	7.290		35.500	1.786	1.770	1.272	0.492	0.492	0.068	7.990		35.500	2.072	2.050	1.265	0.484	0.4
36.000	4.100		36.000	0.332	0.330	1.403	0.560	0.560	0.139	5.040		36.000	0.895	0.890	1.223	0.484	0.484	0.084	6.110		36.000	1.529	1.520	1.253	0.490	0.490	0.066	6.920		36.000	1.915	1.900	1.256	0.486	0.486	0.062	7.620		36.000	2.199	2.180	1.248	0.478	0.4
36.500	3.660		36.500	0.391	0.390	1.353	0.540	0.540	0.119	4.640		36.500	0.994	0.990	1.196	0.473	0.473	0.073	5.730		36.500	1.648	1.640	1.231	0.481	0.481	0.058	6.560		36.500	2.052	2.040	1.241	0.480	0.480	0.056	7.270		36.500	2.349	2.330	1.237	0.474	0.4
37.000	3.230		37.000	0.460	0.460	1.302	0.520	0.520	0.098	4.250		37.000	1.104	1.100	1.169	0.462	0.462	0.063	5.370		37.000	1.788	1.780	1.214	0.475	0.475	0.051	6.200		37.000	2.192	2.180	1.223	0.473	0.473	0.049	6.920		37.000	2.495	2.480	1.222	0.468	0.4
37.500	2.810		37.500	0.541	0.540	1.248	0.498	0.498	0.077	3.880		37.500	1.233	1.230	1.147	0.454	0.454	0.054	5.010		37.500	1.926	1.920	1.193	0.467	0.467	0.043	5.850		37.500	2.340	2.330	1.207	0.467	0.467	0.043	6.570		37.500	2.644	2.630	1.206	0.462	0.4
38.000	2.420		38.000	0.650	0.650	1.208	0.482	0.482	0.061	3.510		38.000	1.362	1.360	1.118	0.442	0.442	0.042	4.670		38.000	2.085	2.080	1.177	0.460	0.460	0.036	5.510		38.000	2.498	2.490	1.190	0.461	0.461	0.036	6.240		38.000	2.812	2.800	1.194	0.457	0.4
38.500	2.040		38.500	0.770	0.770	1.158	0.462	0.462	0.041	3.160		38.500	1.512	1.510	1.093	0.432	0.432	0.032	4.330		38.500	2.245	2.240	1.157	0.452	0.452	0.029	5.180		38.500	2.667	2.660	1.175	0.455	0.455	0.031	5.910		38.500	2.981	2.970	1.180	0.452	0.4
39.000	1.690		39.000	0.921	0.920	1.116	0.446	0.446	0.024	2.830		39.000	1.682	1.680	1.071	0.423	0.423	0.024	4.010		39.000	2.424	2.420	1.140	0.446	0.446	0.022	4.850		39.000	2.836	2.830	1.157	0.448	0.448	0.023	5.580		39.000	3.148	3.140	1.163	0.446	0.4
39.500	1.370	1.370	39.500	1.101	1.100	1.076	0.430	0.430	0.008	2.510	2.511	39.500	1.862	1.860	1.046	0.414	0.414	0.014	3.690	3.690	39.500	2.602	2.600	1.119	0.438	0.438	0.014	4.540		39.500	3.024	3.020	1.142	0.442	0.442	0.018	5.270		39.500	3.337	3.330	1.150	0.441	0.4
40.000	1.080	1.081	40.000	1.311	1.310	1.036	0.414	0.414	-0.008	2.210	2.210	40.000	2.060	2.060	1.022	0.404	0.404	0.004	3.390	3.390	40.000	2.801	2.800	1.102	0.431	0.431	0.007	4.240	4.240	40.000	3.223	3.220	1.128	0.436	0.436	0.012	4.960		40.000	3.525	3.520	1.135	0.435	0.4
40.500	0.840	0.840	40.500	1.570	1.570	1.009	0.403	0.403	-0.018	1.930	1.931	40.500	2.280	2.280	1.000	0.395	0.395	-0.005	3.100	3.100	40.500	3.010	3.010	1.084	0.424	0.424	0.000	3.940	3.939	40.500	3.421	3.420	1.110	0.430	0.430	0.005	4.660	4.661	40.500	3.724	3.720	1.120	0.429	0.4
41.000	0.630	0.630	41.000		1.860	0.979	0.391	0.391	-0.031	1.670	1.670	41.000	2.519	2.520	0.977	0.386	0.386	-0.013	2.830	2.830	41.000	3.239	3.240	1.068	0.418	0.418	-0.006	3.660	3.661	41.000	3.641	3.640	1.096	0.424	0.424	0.000	4.360	4.361	41.000	3.923	3.920	1.103	0.423	0.4
41.500	0.460	0.460	41.500		2.190	0.954	0.381	0.381	-0.040	1.430	1.429	41.500	2.778	2.780	0.955	0.378	0.378	-0.022	2.570	2.570	41.500	3.478	3.480	1.052	0.411	0.411	-0.012	3.390	3.390	41.500	3.868	3.870	1.081	0.418	0.418	-0.006	4.080	4.079	41.500	4.139	4.140	1.089	0.417	0.4
42.000	0.340	0.340	42.000		2.560	0.948	0.378	0.378	-0.043	1.210	1.210	42.000		3.060	0.934	0.369	0.369	-0.031	2.320	2.321	42.000	3.728	3.730	1.035	0.405	0.405	-0.019	3.130	3.130	42.000	4.107	4.110	1.067	0.413	0.413	-0.011	3.810	3.810	42.000	4.368	4.370	1.076	0.412	0.4
42.500	0.240	0.240	42.500		2.970	0.935	0.373	0.373	-0.048	1.020	1.020	42.500		3.370	0.918	0.363	0.363	-0.037	2.090	2.089	42.500	3.995	4.000	1.019	0.399	0.399	-0.025	2.890	2.890	42.500	4.365	4.370	1.055	0.408	0.408	-0.016	3.550	3.550	42.500	4.606	4.610	1.062	0.407	0.4
43.000	0.170	0.171	43.000		3.400	0.933	0.372	0.372	-0.049	0.860	0.861	43.000		3.710	0.907	0.359	0.359	-0.041	1.880	1.880	43.000	4.285	4.290	1.007	0.394	0.394	-0.030	2.650	2.650	43.000	4.624	4.630	1.040	0.403	0.403	-0.022	3.300	3.299	43.000	4.854	4.860	1.049	0.402	0.4
43.500	0.120	0.120	43.500		3.850	0.933	0.372	0.372	-0.049	0.720	0.719	43.500		4.070	0.896	0.354	0.354	-0.045	1.680	1.680	43.500		4.590	0.993	0.389	0.389	-0.035	2.430	2.429	43.500	4.902	4.910	1.028	0.398	0.398	-0.026	3.070	3.071	43.500	5.123	5.130	1.039	0.398	0.3
44.000	0.090	0.089	44.000		4.320	0.948	0.378	0.378	-0.043	0.590	0.590	44.000		4.440	0.883	0.349	0.349	-0.051	1.500	1.500	44.000		4.910	0.982	0.384	0.384	-0.040	2.220	2.221	44.000	5.193	5.200	1.016	0.393	0.393	-0.031	2.840	2.841	44.000	5.392	5.400	1.026	0.393	0.3
44.500	0.070	0.070	44.500		4.800	0.975	0.389	0.389	-0.032	0.490	0.490	44.500		4.840	0.877	0.347	0.347	-0.053	1.340	1.341	44.500		5.250	0.974	0.381	0.381	-0.043	2.020	2.021	44.500		5.500	1.003	0.388	0.388	-0.036	2.630	2.630	44.500	5.679	5.690	1.015	0.389	0.3
45.000	0.060	0.060	45.000		5.290	1.017	0.406	0.406	-0.015	0.410	0.409	45.000		5.260	0.876	0.346	0.346	-0.053	1.190	1.189	45.000		5.600	0.965	0.377	0.377	-0.047	1.840	1.840	45.000		5.820	0.994	0.385	0.385	-0.040	2.430	2.430	45.000	5.977	5.990	1.005	0.385	0.3
45.500	0.050	0.051	45.500		5.780	1.055	0.421	0.421	0.000	0.340	0.339	45.500		5.690	0.874	0.346	0.346	-0.054	1.050	1.050	45.500		5.960	0.955	0.373	0.373	-0.050	1.680	1.680	45.500		6.160	0.987	0.382	0.382	-0.042	2.240	2.239	45.500		6.300	0.995	0.381	0.3
46.000	0.040	0.040	46.000		6.270	1.078	0.430	0.430	0.009	0.290	0.290	46.000		6.140	0.882	0.349	0.349	-0.051	0.930	0.930	46.000		6.340	0.949	0.371	0.371	-0.053	1.530	1.529	46.000		6.510	0.980	0.379	0.379	-0.045	2.060	2.060	46.000		6.620	0.986	0.378	0.3
46.500	0.040	0.040	46.500		6.770	1.139	0.455	0.455	0.034	0.240	0.240	46.500		6.590	0.882	0.349	0.349	-0.051	0.830	0.831	46.500		6.740	0.947	0.370	0.370	-0.053	1.390	1.390	46.500		6.870	0.973	0.377	0.377	-0.047	1.900	1.901	46.500		6.960	0.979	0.375	0.3
47.000	0.030	0.029	47.000		7.260	1.147	0.458	0.458	0.037	0.210	0.210	47.000		7.060	0.895	0.354	0.354	-0.046	0.740	0.740	47.000		7.150	0.945	0.370	0.370	-0.054	1.260	1.260	47.000		7.240	0.967	0.374	0.374	-0.050	1.750	1.750	47.000		7.310	0.973	0.373	0.3
47.500	0.030	0.030	47.500		7.760	1.208	0.482	0.482	0.061	0.180	0.180	47.500		7.530	0.903	0.357	0.357	-0.043	0.660	0.661	47.500		7.570	0.945	0.370	0.370	-0.054	1.140	1.140	47.500		7.620	0.961	0.372	0.372	-0.052	1.610	1.609	47.500		7.670	0.967	0.370	0.3
48.000	0.020	0.020	48.000		8.250	1.193	0.476	0.476	0.055	0.150	0.151	48.000		8.000	0.907	0.359	0.359	-0.041	0.580	0.580	48.000		7.990	0.940	0.368	0.368	-0.056	1.040	1.039	48.000		8.010	0.959	0.371	0.371	-0.053	1.480	1.480	48.000		8.040	0.962	0.369	0.3
48.500	0.020	0.020	48.500		8.750	1.254	0.500	0.500	0.079	0.130	0.130	48.500		8.480	0.917	0.362	0.362	-0.037	0.520	0.519	48.500		8.430	0.942	0.369	0.369	-0.055	0.940	0.941	48.500		8.420	0.955	0.370	0.370	-0.055	1.360	1.360	48.500		8.420	0.958	0.367	0.3
49.000	0.020	0.021	49.000		9.250	1.315	0.525	0.525	0.104	0.120	0.120	49.000		8.970	0.938	0.371	0.371	-0.029	0.470	0.471	49.000		8.880	0.948	0.371	0.371	-0.053	0.860	0.859	49.000		8.840	0.955	0.370	0.370	-0.055	1.250	1.250	49.000		8.810	0.954	0.365	0.3



Option Grid (Search) – Core Features

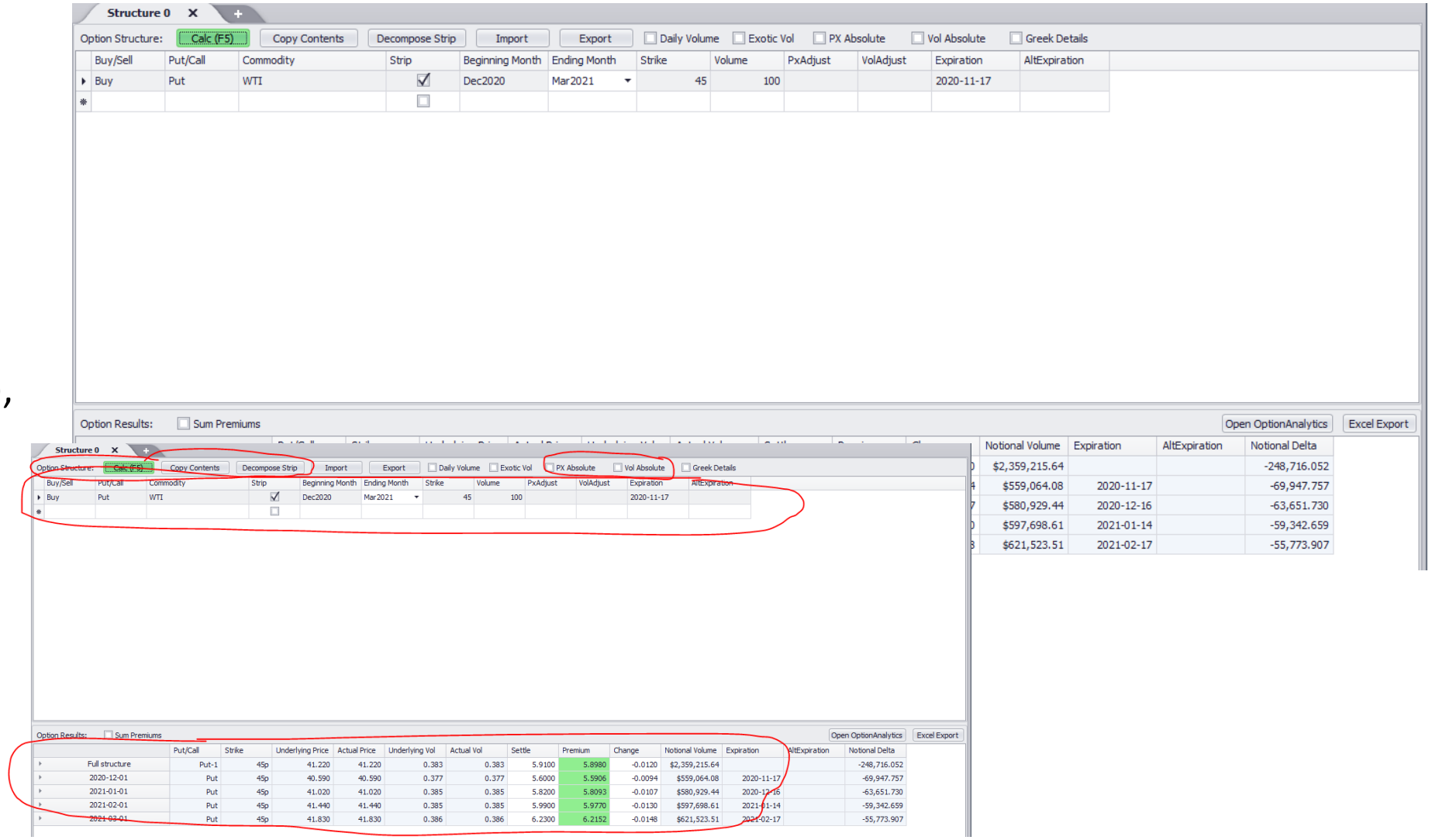
- Above the Option Grid screen is a search box whereby a user can enter in a formula like we did with Option String.
- This produces the value of that option.



	Oct20(V20) | $39.77 | 0.00									Nov20(X20) | $40.15 | 0.00									Dec20(Z20) | $40.59 | 0.00									Jan21(F21) | $41.02 | 0.00									Feb21(G21) | $41.44 | 0.00						
	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpVol	VolDiff	CSettle	Calls	Strike	Puts	PSettle	GammaBE	SetVol	ImpV
30.000	9.850		30.000	0.083	0.080	2.194	0.876	0.876	0.455	10.440		30.000	0.301	0.290	1.644	0.650	0.650	0.250	11.170		30.000	0.601	0.590	1.531	0.599	0.599	0.175	11.820		30.000	0.831	0.800	1.486	0.575	0.575	0.151	12.400		30.000	1.001	0.960	1.440	0.552	0.5
30.500	9.360		30.500	0.094	0.090	2.133	0.851	0.851	0.430	9.970		30.500	0.330	0.320	1.609	0.636	0.636	0.237	10.720		30.500	0.650	0.630	1.506	0.589	0.589	0.165	11.380		30.500	0.890	0.850	1.463	0.566	0.566	0.142	11.970		30.500	1.069	1.030	1.421	0.544	0.5
31.000	8.870		31.000	0.104	0.100	2.064	0.824	0.824	0.403	9.490		31.000	0.350	0.350	1.559	0.616	0.616	0.217	10.270		31.000	0.700	0.680	1.479	0.578	0.578	0.154	10.940		31.000	0.948	0.930	1.438	0.557	0.557	0.132	11.550		31.000	1.147	1.110	1.405	0.538	0.5
31.500	8.380		31.500	0.113	0.110	1.988	0.793	0.793	0.372	9.030		31.500	0.390	0.380	1.532	0.606	0.606	0.206	9.820		31.500	0.748	0.740	1.448	0.566	0.566	0.143	10.510		31.500	1.017	1.000	1.417	0.549	0.549	0.124	11.130		31.500	1.225	1.190	1.387	0.531	0.5
32.000	7.890		32.000	0.122	0.120	1.911	0.763	0.763	0.342	8.560		32.000	0.419	0.410	1.488	0.588	0.588	0.188	9.380		32.000	0.808	0.800	1.423	0.557	0.557	0.133	10.090		32.000	1.094	1.070	1.399	0.541	0.541	0.117	10.720		32.000	1.314	1.280	1.372	0.526	0.5
32.500	7.400		32.500	0.133	0.130	1.835	0.732	0.732	0.311	8.100		32.500	0.459	0.450	1.453	0.575	0.575	0.175	8.950		32.500	0.876	0.860	1.401	0.548	0.548	0.124	9.670		32.500	1.174	1.150	1.379	0.534	0.534	0.109	10.310		32.500	1.403	1.370	1.355	0.519	0.5
33.000	6.920		33.000	0.153	0.150	1.778	0.710	0.710	0.288	7.640		33.000	0.498	0.490	1.413	0.559	0.559	0.159	8.520		33.000	0.946	0.930	1.376	0.538	0.538	0.114	9.260		33.000	1.262	1.240	1.361	0.527	0.527	0.103	9.910		33.000	1.500	1.470	1.340	0.513	0.5
33.500	6.440		33.500	0.171	0.170	1.713	0.684	0.684	0.262	7.190		33.500	0.547	0.540	1.380	0.546	0.546	0.146	8.100		33.500	1.024	1.010	1.354	0.529	0.529	0.106	8.850		33.500	1.352	1.330	1.342	0.519	0.519	0.095	9.520		33.500	1.608	1.580	1.326	0.508	0.5
34.000	5.960		34.000	0.192	0.190	1.644	0.656	0.656	0.235	6.740		34.000	0.597	0.590	1.343	0.531	0.531	0.131	7.690		34.000	1.113	1.100	1.334	0.522	0.522	0.098	8.450		34.000	1.449	1.430	1.323	0.512	0.512	0.088	9.120		34.000	1.706	1.680	1.308	0.501	0.5
34.500	5.480		34.500	0.211	0.210	1.567	0.626	0.626	0.204	6.310		34.500	0.666	0.660	1.317	0.521	0.521	0.121	7.280		34.500	1.202	1.190	1.312	0.513	0.513	0.089	8.060		34.500	1.558	1.540	1.307	0.506	0.506	0.082	8.740		34.500	1.826	1.800	1.295	0.496	0.4
35.000	5.020		35.000	0.252	0.250	1.525	0.609	0.609	0.188	5.880		35.000	0.736	0.730	1.286	0.509	0.509	0.109	6.880		35.000	1.300	1.290	1.290	0.505	0.505	0.081	7.670		35.000	1.666	1.650	1.288	0.499	0.499	0.074	8.360		35.000	1.943	1.920	1.279	0.490	0.4
35.500	4.550		35.500	0.281	0.280	1.453	0.580	0.580	0.159	5.450		35.500	0.805	0.800	1.250	0.494	0.494	0.095	6.490		35.500	1.409	1.400	1.271	0.497	0.497	0.073	7.290		35.500	1.786	1.770	1.272	0.492	0.492	0.068	7.990		35.500	2.072	2.050	1.265	0.484	0.4
36.000	4.100		36.000	0.332	0.330	1.403	0.560	0.560	0.139	5.040		36.000	0.895	0.890	1.223	0.484	0.484	0.084	6.110		36.000	1.529	1.520	1.253	0.490	0.490	0.066	6.920		36.000	1.915	1.900	1.256	0.486	0.486	0.062	7.620		36.000	2.199	2.180	1.248	0.478	0.4
36.500	3.660		36.500	0.391	0.390	1.353	0.540	0.540	0.119	4.640		36.500	0.994	0.990	1.196	0.473	0.473	0.073	5.730		36.500	1.648	1.640	1.231	0.481	0.481	0.058	6.560		36.500	2.052	2.040	1.241	0.480	0.480	0.056	7.270		36.500	2.349	2.330	1.237	0.474	0.4
37.000	3.230		37.000	0.460	0.460	1.302	0.520	0.520	0.098	4.250		37.000	1.104	1.100	1.169	0.462	0.462	0.063	5.370		37.000	1.788	1.780	1.214	0.475	0.475	0.051	6.200		37.000	2.192	2.180	1.223	0.473	0.473	0.049	6.920		37.000	2.495	2.480	1.222	0.468	0.4
37.500	2.810		37.500	0.541	0.540	1.248	0.498	0.498	0.077	3.880		37.500	1.233	1.230	1.147	0.454	0.454	0.054	5.010		37.500	1.926	1.920	1.193	0.467	0.467	0.043	5.850		37.500	2.340	2.330	1.207	0.467	0.467	0.043	6.570		37.500	2.644	2.630	1.206	0.462	0.4
38.000	2.420		38.000	0.650	0.650	1.208	0.482	0.482	0.061	3.510		38.000	1.362	1.360	1.118	0.442	0.442	0.042	4.670		38.000	2.085	2.080	1.177	0.460	0.460	0.036	5.510		38.000	2.498	2.490	1.190	0.461	0.461	0.036	6.240		38.000	2.812	2.800	1.194	0.457	0.4
38.500	2.040		38.500	0.770	0.770	1.158	0.462	0.462	0.041	3.160		38.500	1.512	1.510	1.093	0.432	0.432	0.032	4.330		38.500	2.245	2.240	1.157	0.452	0.452	0.029	5.180		38.500	2.667	2.660	1.175	0.455	0.455	0.031	5.910		38.500	2.981	2.970	1.180	0.452	0.4
39.000	1.690		39.000	0.921	0.920	1.116	0.446	0.446	0.024	2.830		39.000	1.682	1.680	1.071	0.423	0.423	0.024	4.010		39.000	2.424	2.420	1.140	0.446	0.446	0.022	4.850		39.000	2.836	2.830	1.157	0.448	0.448	0.023	5.580		39.000	3.148	3.140	1.163	0.446	0.4
39.500	1.370	1.370	39.500	1.101	1.100	1.076	0.430	0.430	0.008	2.510	2.511	39.500	1.862	1.860	1.046	0.414	0.414	0.014	3.690	3.690	39.500	2.602	2.600	1.119	0.438	0.438	0.014	4.540		39.500	3.024	3.020	1.142	0.442	0.442	0.018	5.270		39.500	3.337	3.330	1.150	0.441	0.4
40.000	1.080	1.081	40.000	1.311	1.310	1.036	0.414	0.414	-0.008	2.210	2.210	40.000	2.060	2.060	1.022	0.404	0.404	0.004	3.390	3.390	40.000	2.801	2.800	1.102	0.431	0.431	0.007	4.240	4.240	40.000	3.223	3.220	1.128	0.436	0.436	0.012	4.960		40.000	3.525	3.520	1.135	0.435	0.4
40.500	0.840	0.840	40.500	1.570	1.570	1.009	0.403	0.403	-0.018	1.930	1.931	40.500	2.280	2.280	1.000	0.395	0.395	-0.005	3.100	3.100	40.500	3.010	3.010	1.084	0.424	0.424	0.000	3.940	3.939	40.500	3.421	3.420	1.110	0.430	0.430	0.005	4.660	4.661	40.500	3.724	3.720	1.120	0.429	0.4
41.000	0.630	0.630	41.000		1.860	0.979	0.391	0.391	-0.031	1.670	1.670	41.000	2.519	2.520	0.977	0.386	0.386	-0.013	2.830	2.830	41.000	3.239	3.240	1.068	0.418	0.418	-0.006	3.660	3.661	41.000	3.641	3.640	1.096	0.424	0.424	0.000	4.360	4.361	41.000	3.923	3.920	1.103	0.423	0.4
41.500	0.460	0.460	41.500		2.190	0.954	0.381	0.381	-0.040	1.430	1.429	41.500	2.778	2.780	0.955	0.378	0.378	-0.022	2.570	2.570	41.500	3.478	3.480	1.052	0.411	0.411	-0.012	3.390	3.390	41.500	3.868	3.870	1.081	0.418	0.418	-0.006	4.080	4.079	41.500	4.139	4.140	1.089	0.417	0.4
42.000	0.340	0.340	42.000		2.560	0.948	0.378	0.378	-0.043	1.210	1.210	42.000		3.060	0.934	0.369	0.369	-0.031	2.320	2.321	42.000	3.728	3.730	1.035	0.405	0.405	-0.019	3.130	3.130	42.000	4.107	4.110	1.067	0.413	0.413	-0.011	3.810	3.810	42.000	4.368	4.370	1.076	0.412	0.4
42.500	0.240	0.240	42.500		2.970	0.935	0.373	0.373	-0.048	1.020	1.020	42.500		3.370	0.918	0.363	0.363	-0.037	2.090	2.089	42.500	3.995	4.000	1.019	0.399	0.399	-0.025	2.890	2.890	42.500	4.365	4.370	1.055	0.408	0.408	-0.016	3.550	3.550	42.500	4.606	4.610	1.062	0.407	0.4
43.000	0.170	0.171	43.000		3.400	0.933	0.372	0.372	-0.049	0.860	0.861	43.000		3.710	0.907	0.359	0.359	-0.041	1.880	1.880	43.000	4.285	4.290	1.007	0.394	0.394	-0.030	2.650	2.650	43.000	4.624	4.630	1.040	0.403	0.403	-0.022	3.300	3.299	43.000	4.854	4.860	1.049	0.402	0.4
43.500	0.120	0.120	43.500		3.850	0.933	0.372	0.372	-0.049	0.720	0.719	43.500		4.070	0.896	0.354	0.354	-0.045	1.680	1.680	43.500		4.590	0.993	0.389	0.389	-0.035	2.430	2.429	43.500	4.902	4.910	1.028	0.398	0.398	-0.026	3.070	3.071	43.500	5.123	5.130	1.039	0.398	0.3
44.000	0.090	0.089	44.000		4.320	0.948	0.378	0.378	-0.043	0.590	0.590	44.000		4.440	0.883	0.349	0.349	-0.051	1.500	1.500	44.000		4.910	0.982	0.384	0.384	-0.040	2.220	2.221	44.000	5.193	5.200	1.016	0.393	0.393	-0.031	2.840	2.841	44.000	5.392	5.400	1.026	0.393	0.3
44.500	0.070	0.070	44.500		4.800	0.975	0.389	0.389	-0.032	0.490	0.490	44.500		4.840	0.877	0.347	0.347	-0.053	1.340	1.341	44.500		5.250	0.974	0.381	0.381	-0.043	2.020	2.021	44.500		5.500	1.003	0.388	0.388	-0.036	2.630	2.630	44.500	5.679	5.690	1.015	0.389	0.3
45.000	0.060	0.060	45.000		5.290	1.017	0.406	0.406	-0.019	0.410	0.409	45.000		5.260	0.876	0.346	0.346	-0.053	1.190	1.189	45.000		5.600	0.965	0.377	0.377	-0.047	1.840	1.840	45.000		5.820	0.994	0.385	0.385	-0.040	2.430	2.430	45.000	5.977	5.990	1.005	0.385	0.3
45.500	0.050	0.051	45.500		5.780	1.055	0.421	0.421	0.000	0.340	0.339	45.500		5.690	0.874	0.346	0.346	-0.054	1.050	1.050	45.500		5.960	0.955	0.373	0.373	-0.050	1.680	1.680	45.500		6.160	0.987	0.382	0.382	-0.042	2.240	2.239	45.500		6.300	0.995	0.381	0.3
46.000	0.040	0.040	46.000		6.270	1.078	0.430	0.430	0.009	0.290	0.290	46.000		6.140	0.882	0.349	0.349	-0.051	0.930	0.930	46.000		6.340	0.949	0.371	0.371	-0.053	1.530	1.529	46.000		6.510	0.980	0.379	0.379	-0.045	2.060	2.060	46.000		6.620	0.986	0.378	0.3
46.500	0.040	0.040	46.500		6.770	1.139	0.455	0.455	0.034	0.240	0.240	46.500		6.590	0.882	0.349	0.349	-0.051	0.830	0.831	46.500		6.740	0.947	0.370	0.370	-0.053	1.390	1.390	46.500		6.870	0.973	0.377	0.377	-0.047	1.900	1.901	46.500		6.960	0.979	0.375	0.3
47.000	0.030	0.029	47.000		7.260	1.147	0.458	0.458	0.037	0.210	0.210	47.000		7.060	0.895	0.354	0.354	-0.046	0.740	0.740	47.000		7.150	0.945	0.370	0.370	-0.054	1.260	1.260	47.000		7.240	0.967	0.374	0.374	-0.050	1.750	1.750	47.000		7.310	0.973	0.373	0.3
47.500	0.030	0.030	47.500		7.760	1.208	0.482	0.482	0.061	0.180	0.180	47.500		7.530	0.903	0.357	0.357	-0.043	0.660	0.661	47.500		7.570	0.945	0.370	0.370	-0.054	1.140	1.140	47.500		7.620	0.961	0.372	0.372	-0.052	1.610	1.609	47.500		7.670	0.967	0.370	0.3
48.000	0.020	0.020	48.000		8.250	1.193	0.476	0.476	0.055	0.150	0.151	48.000		8.000	0.907	0.359	0.359	-0.041	0.580	0.580	48.000		7.990	0.940	0.368	0.368	-0.056	1.040	1.039	48.000		8.010	0.959	0.371	0.371	-0.053	1.480	1.480	48.000		8.040	0.962	0.369	0.3
48.500	0.020	0.020	48.500		8.750	1.254	0.500	0.500	0.079	0.130	0.130	48.500		8.480	0.917	0.362	0.362	-0.037	0.520	0.519	48.500		8.430	0.942	0.369	0.369	-0.055	0.940	0.941	48.500		8.420	0.955	0.370	0.370	-0.055	1.360	1.360	48.500		8.420	0.958	0.367	0.3
49.000	0.020	0.021	49.000		9.250	1.315	0.525	0.525	0.104	0.120	0.120	49.000		8.970	0.938	0.371	0.371	-0.029	0.470	0.471	49.000		8.880	0.948	0.371	0.371	-0.053	0.860	0.859	49.000		8.840	0.955	0.370	0.370	-0.055	1.250	1.250	49.000		8.810	0.954	0.365	0.3

Pricer (Options) – Core Features

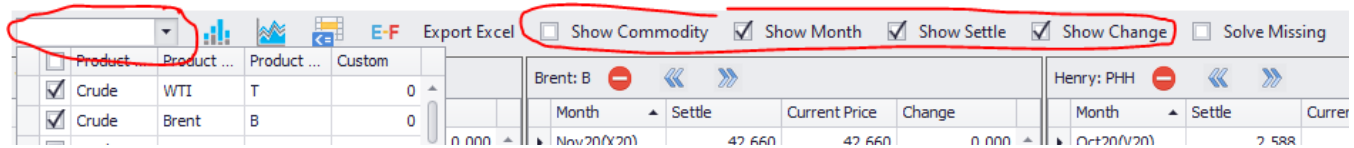
- Screen whereby a user can have multiple legs
- Both single and strips (i.e. multiple months)
- Ability to see pricing by leg
- See put, call, strike, underlying price, actual price (if live or adjusted), Underlying volatility, Adjusted volatility (if they did Vol Adjust), Option settle, Premium, Change (from settle), notional volume, notional delta.



Futures – Core Features

- By commodity (drop down menu)
- Add months
- Ability to show multiple commodities
- Should show commodity, month, settle, last, change

MONTH	OPEN	HIGH	LOW	LAST	CHANGE	SETTLE	ESTIMATED VOLUME	PRIOR DAY OPEN INTEREST
OCT 20	2.483	2.605	2.431	2.579	+.101	2.588	172,758	211,026
NOV 20	2.924	3.002	2.882	2.981	+.060	2.987	92,584	204,144
DEC 20	3.257	3.319	3.229	3.276	+.017	3.276	53,698	103,852
JAN 21	3.374	3.433	3.346	3.390	+.015	3.389	38,985	127,740
FEB 21	3.331	3.389	3.306	3.352	+.017	3.350	16,589	48,657
MAR 21	3.190	3.245	3.171	3.216B	+.016	3.217	18,371	98,260
APR 21	2.840	2.872	2.836	2.849	+.002	2.852	9,831	78,605
MAY 21	2.794	2.822	2.791	2.806	+.003	2.806	6,041	46,389
JUN 21	2.826	2.846	2.816	2.830A	+.005	2.830	3,305	27,368
JLY 21	2.847	2.876	2.847	2.860	+.005	2.860	1,486	20,167
AUG 21	2.860	2.882	2.855A	2.867	+.005	2.867	1,050	20,973



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Month	Settle	Current Price	Change
Oct20(V20)	39.770	39.770	0.000
Nov20(X20)	40.150	40.150	0.000
Dec20(Z20)	40.590	40.590	0.000
Jan21(F21)	41.020	41.020	0.000
Feb21(G21)	41.440	41.440	0.000
Mar21(H21)	41.830	41.830	0.000
Apr21(J21)	42.200	42.200	0.000
May21(K21)	42.530	42.530	0.000
Jun21(M21)	42.840	42.840	0.000
Jul21(N21)	43.090	43.090	0.000
Aug21(Q21)	43.300	43.300	0.000
Sep21(U21)	43.490	43.490	0.000
Oct21(V21)	43.660	43.660	0.000
Nov21(X21)	43.810	43.810	0.000
Dec21(Z21)	43.940	43.940	0.000
Jan22(F22)	44.040	44.040	0.000
Feb22(G22)	44.150	44.150	0.000
Mar22(H22)	44.270	44.270	0.000
Apr22(J22)	44.390	44.390	0.000
May22(K22)	44.510	44.510	0.000

Brent: B

Month	Settle	Current Price	Change
Nov20(X20)	42.660	42.660	0.000
Dec20(Z20)	43.180	43.180	0.000
Jan21(F21)	43.650	43.650	0.000
Feb21(G21)	44.110	44.110	0.000
Mar21(H21)	44.550	44.550	0.000
Apr21(J21)	44.980	44.980	0.000
May21(K21)	45.370	45.370	0.000
Jun21(M21)	45.700	45.700	0.000
Jul21(N21)	45.970	45.970	0.000
Aug21(Q21)	46.240	46.240	0.000
Sep21(U21)	46.480	46.480	0.000
Oct21(V21)	46.700	46.700	0.000
Nov21(X21)	46.870	46.870	0.000
Dec21(Z21)	47.020	47.020	0.000
Jan22(F22)	47.170	47.170	0.000
Feb22(G22)	47.350	47.350	0.000
Mar22(H22)	47.570	47.570	0.000
Apr22(J22)	47.770	47.770	0.000
May22(K22)	47.960	47.960	0.000
Jun22(M22)	48.140	48.140	0.000

User Profile– Core Features

- Screen to adjust their password
- Screen to see their username name
- Name and company info

Admin Screen – Core Features

- Screen 1: Account management
  - Ability to setup a user (when new user is added, an email is sent out with their username and password)
  - Ability to select what commodities they can see (drop down checkbox)
  - Ability to assign what company they work for
- Screen 2: Access log
  - See what users have accessed with date and time
- Screen 3: Structure log
  - See a list of what user has logged in, what time and what they have priced, with the value